SAMS Kernel

What has run on real hardware, and how to try the rest

Version 1.16.0 · 28 October 2026
For a TI-99/4A with an F18A video card and a SAMS memory card
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Everything in this project is verified in a simulator (a TMS9900, the F18A, SAMS, a keyboard, joysticks, TI's own console and disk controller ROMs, a model of the sound chip). That finds almost every programming mistake and none of the hardware's surprises. This list is the honest boundary, in the order I would walk it on a real machine or in js99er (F18A and SAMS on). Each step says what to do, what you should see, and what to tell me if you do not.

Run on real hardware before (by Press and The Lantern Coast, the kernel's ancestors)

SAMS paging with a 1M-or-smaller card, the keyboard decode, the F18A 80-column text screen and colour roles, files through TI's disk controller, the FinalGROM "8" image format and bank switching. **Not** run on real hardware in this form: the loader as rewritten, the module system, the page pool, and everything below. js99er with a 16M SAMS has run the kernel (which found the QUIT bug in 1.3.0, fixed in 1.3.1).

Routes. Steps 1-8 run prebuilt images (`images/`), which need a **[PC]** only to have been made: copy them to the cartridge and the disk and you can do them with no PC. Step 9 has two forms below: **[PC]** builds `tests/apps/files2`; **[TI]** runs `COPY/A99` from the workbench disk (the same test, two files open at once). Steps 10 and 11 are **[TI]** by nature (the workbench itself).

The walk

#	Do	You should see	If not, tell me
1	Load `images/hello8.bin`. Type. FCTN =	a blue screen, a title bar, your typing echoed; FCTN = returns to the console's title screen with the display normal	which message (a refusal screen names its reason), a blank screen, or garbled colours
2	`images/skdiag8.bin`, CTRL T	eight rows, all **PASS**; the first shows the page count (128 for 512K, 256 for 1M, 512 / 1024 / ... / 4096 for bigger cards); a disk in drive 1 for the last row	which row failed; the step and error on the files row
3	Leave diag with FCTN = (**with a card above 1M if you have one**)	the console title screen; the card's mapping must be off	a hang or blank screen at the reset
4	In diag, press keys and chords	each shows its code and modifiers; CTRL 3 cycles the colour schemes	a chord with the wrong code (name it)
5	`images/videomodes8.bin`, keys `1-9 0 A-E`; FCTN =	fifteen screens, one for each display mode and option; each matches `docs/VIDEO-MODES.md`	which mode, and what it looked like; PICO9918 or F18A, and its firmware
6	`images/music8.bin`: `1`, SPACE, SPACE, `S`, `2`, `B`, `W`	the tune (three voices and a beat, about 16 s, looping), silent while paused and then the same note, an A (440 Hz), a beep, and the tune playing through a two-second "work" loop	the tune is too fast or slow, a wrong note, a hiss or click, or the noise voice too loud (a PAL console needs `"fps": 50`)
7	`images/joystick8.bin`, each joystick in each direction, and fire	the matching label goes bold; the fire count rises once a press; `1`, `2`, `3` make joystick 1, 2 or both type keys (the "Keys typed" count and the last code change; UP, DOWN, LEFT, RIGHT, ENTER are `C5 D8 D3 C4 0D`)	the wrong direction or joystick, a key that sticks, ghosting with a key held
8	`images/diskbrowser8.bin` with a disk in DSK1	the catalog; ENTER on a text file shows it; scroll; FCTN 9 back	a hang, a wrong size or type, a garbled name

9	Several files, **[TI]**: in the workbench (the disk has a sample `SRC`), N `DSK1.COPY/A99` ENTER, A, R, ENTER; the disk then holds `DST`. **[PC]**: build `tests/apps/files2` (`python3 tools/skbuild.py tests/apps/files2`) and make a disk from two text files named `src` and `src2` (`python3 tools/skdisk.py d.dsk src src2`), then load `build/files2/files28.bin` with that disk in DSK1	nothing on the screen; afterwards the disk holds `DST`, a copy of `SRC` (check it in your usual disk manager)	the disk controller or TIPI in use; whether `DST` is complete
10	`images/workbench8.bin` with `images/workbench.dsk` in DSK1: N `DSK1.HELLO/A99` ENTER, A, then R	"Assembled: 194 bytes at >C000", then the hello program; **time the assembly with a watch and tell me** (the simulator ran 1.9 million instructions); then restart (hello has no way back) and `BEEP/A99`: A, R, keys 1-8 play a scale, FCTN 9 returns to the workbench	the time; any error list; a hang
11	In the workbench: E on `HELLO/A99`, type a line, CTRL S, CTRL A	the edit saves and re-assembles; a mistyped mnemonic is reported with its line	an editor key that does the wrong thing
12	Cards above 1M, if you have one: step 2 again, then 6 and 10	the page count is right and everything else as before	the first thing that differs
12a	In the workbench: N `DSK1.ASK/A99` ENTER, A, R; type `ann` ENTER, `bob` ENTER, then FCTN E	`Hello, ANN`, `Hello, BOB`; FCTN E puts BOB back in the field; the arrows, FCTN 1, CTRL H and FCTN 2 edit; FCTN 9 returns	a key that does the wrong thing (name it), a field that does not redraw cleanly

TIPI steps (if you have one; each is **[TI]**):

#	Do	You should see	If not, tell me
13	`images/diskbrowser8.bin` with `DSK1.` mapped to a directory on the TIPI	the directory's catalog; a text file opens	whether the catalog request is answered at all
14	the workbench on the TIPI: a source in the mapped directory (or `PI.` paths), N, A, R	assembles and runs, as from a disk; includes found	the first failing name, and its error
15	`COPY/A99` with `SRC` on `DSK1.` (TIPI) and, if you also have a TI disk controller, the copy going between them	`DST` appears; with both cards present nothing hangs, whichever device is used first	which direction fails
16	`CLOCK/A99` from the workbench disk, ENTER	today's date and the time	"no clock" with a TIPI attached, or a wrong date

What each unverified claim rests on

Claim	Rests on	Where it could be wrong
a 12-bit SAMS register is the page with its bytes swapped (even byte the low 8 bits)	js99er's `ams.js`	a real card above 1M might lay the odd byte out differently
joysticks are keyboard columns 6 and 7, rows fire / left / right / down / up	my knowledge of the console's scan, and the simulator's model of it	the row order, or a joystick that needs a different column select
three files at once through the controller's three buffers; the controller finds a file by its PAB's address	TI's own disk controller ROM, running in the simulator (it did so correctly)	TIPI, CF7+ and other devices may differ
the music player writes the right bytes at the right frames	tests against a model of the sound chip	how it sounds; the noise model is an approximation
the workbench's assembler is right	byte-identical to xas99 for every example and two probe sources	an instruction or directive the examples do not use
TIPI: the device names, the clock record, the power-up of every card, includes in the main	the TIPI wiki as Press's design used it (Press's file speed was measured on a TIPI); a model of a TIPI	a real TIPI's DSR, its open-file limit and its catalog

power-up of every card, includes in the main file's directory	speed was measured on a TIPI); a model of a TIPI in the simulator	file limit and its catalog
the page tests at power-up take about a second on a 16M card	0.25 million instructions in the simulator	an estimate; I have not timed it

How to report

A short note per step is enough: the step number, what you saw, the machine (console, F18A or PICO9918 with its firmware, SAMS size and brand, disk controller or TIPI), and for emulators the version and settings. A photograph of a refusal screen or the self-test is the best evidence there is.